

Alpilla 600 MW Combined Cycle Power Plant Topographic Survey and Site Setting-Out Report

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Rev	Purpose	Date	Description	Prepared	Checked	Approved
A	IFR	2026-11-20	First issue for Owner review	EPCE	EPC civil lead	EPC engineering manager

1. Purpose and scope

This report presents the topographic survey of the site, the control network and benchmarks for setting out, the check of the Owner's levelled platform and the levels of the temporary construction area TP-A1, and fixes the setting-out rules for the works. Survey by Land survey subcontractor (role-play), point listing ALP-SUR-TOP-001 Rev 0 received 2026-11-13 (SRC-EPCE-0011): 291 points - 4 control pillars, 3 benchmarks, 271 spot heights, 13 existing features.

2. Reference systems

Item	Value	Remarks
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m m	TUREF (ITRF96), south-west plot corner
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site
Scale factor plant grid to UTM	0.999612	combined UTM point scale and height factor
Height datum	TUDKA-99 orthometric heights (m a.s.l.)	
Site grade level	15 m a.s.l.	

Plant grid (E, N) in metres, origin at the south-west plot corner, plant north = true north (ALP-OWN-GEN-001). UTM = origin + k x R(rotation) x (E, N). All design drawings use plant grid coordinates; UTM coordinates are used for the authority submissions and the external connections.

3. Control network and benchmarks

Primary control pillars (GNSS static observations) with the residual of the transformation from plant grid to UTM against the measured coordinates:

Point	E plant	N plant	E UTM 35N	N UTM 35N	Height	Residual mm
CP-01	15.000	12.000	541265.052	4532491.905	15.046	11
CP-02	392.000	12.000	541641.906	4532489.755	15.018	6
CP-03	392.000	293.000	541643.520	4532770.638	15.071	1
CP-04	15.000	293.000	541266.673	4532772.793	15.083	3

Benchmark	E plant	N plant	Height m	Description
BM-01	30.0	296.0	15.312	Site benchmark, brass bolt in concrete block; levelled from national benchmark NB-35-1147 (+18.412 m TUDKA-99), closure 3 mm
BM-02	381.0	15.0	15.108	Site benchmark, brass bolt in concrete block; levelled loop BM-01 - BM-02, closure 2 mm
BM-03	90.0	590.0	15.466	Temporary area benchmark, steel pin in concrete; from BM-01, closure 4 mm

4. Platform levels

Plot (208 spot heights on a 25 m grid, stream channel strip excluded): levels 14.70 to 15.17 m, mean deviation from the platform level +15.00 m -66 mm, standard deviation 76 mm; 157 of 208 points (75%) within +0.10 m. Trimming to +15.00 m: cut 747 m3, fill 9,324 m3 (grid cell method, 25 x 25 m cells).

Point	E	N	Level m	Deviation mm	Area
SP-0001	25	0	14.716	-284	south edge (setback fence)
SP-0002	25	25	14.858	-142	platform
SP-0003	25	50	14.848	-152	platform

Point	E	N	Level m	Deviation mm	Area
SP-0005	25	100	14.867	-133	platform
SP-0006	25	125	14.895	-105	platform
SP-0007	25	150	14.826	-174	platform
SP-0014	50	0	14.717	-283	south edge (setback fence)
SP-0015	50	25	14.897	-103	platform
SP-0017	50	75	14.881	-119	platform
SP-0027	75	0	14.819	-181	south edge (setback fence)
SP-0029	75	50	14.863	-137	platform
SP-0040	100	0	14.766	-234	south edge (setback fence)
SP-0042	100	50	14.889	-111	platform
SP-0043	100	75	14.874	-126	platform
SP-0045	100	125	14.897	-103	platform
SP-0053	125	0	14.756	-244	south edge (setback fence)
SP-0054	125	25	14.871	-129	platform
SP-0056	125	75	14.895	-105	platform
SP-0057	125	100	14.879	-121	platform
SP-0066	150	0	14.738	-262	south edge (setback fence)
SP-0067	150	25	14.884	-116	platform
SP-0068	150	50	14.866	-134	platform
SP-0069	150	75	14.873	-127	platform
SP-0079	175	0	14.721	-279	south edge (setback fence)
SP-0080	175	25	14.894	-106	platform
SP-0083	175	100	14.893	-107	platform
SP-0092	200	0	14.790	-210	south edge (setback fence)
SP-0093	200	25	14.881	-119	platform
SP-0094	200	50	14.885	-115	platform
SP-0096	200	100	14.890	-110	platform
SP-0105	225	0	14.715	-285	south edge (setback fence)
SP-0106	225	25	14.866	-134	platform
SP-0118	250	0	14.782	-218	south edge (setback fence)
SP-0121	250	75	14.875	-125	platform
SP-0131	275	0	14.750	-250	south edge (setback fence)
SP-0132	275	25	14.840	-160	platform
SP-0144	300	0	14.753	-247	south edge (setback fence)
SP-0146	300	50	15.168	+168	old warehouse foundations
SP-0148	300	100	14.880	-120	platform
SP-0157	325	0	14.748	-252	south edge (setback fence)
SP-0158	325	25	15.136	+136	old warehouse foundations
SP-0160	325	75	14.858	-142	platform
SP-0170	350	0	14.703	-297	south edge (setback fence)
SP-0172	350	50	15.163	+163	old warehouse foundations
SP-0174	350	100	14.895	-105	platform
SP-0183	375	0	14.732	-268	south edge (setback fence)
SP-0185	375	50	14.849	-151	platform
SP-0187	375	100	14.892	-108	platform
SP-0188	375	125	14.899	-101	platform
SP-0196	400	0	14.775	-225	south edge (setback fence)
SP-0198	400	50	14.879	-121	platform

Temporary construction area TP-A1 (63 spot heights, 30 x 35 m grid): levels 13.75 to 15.76 m, rising to the north. Balanced platform level +14.79 m (cut = fill 14,320 m3); the laydown and fabrication areas are levelled to +14.79 m with a 1 % fall to the south-west drainage ditch and restored after Taking-Over (TQ-012).

5. Existing features

Point	E	N	Level m	Feature
FT-01	2	0	13.02	stream channel bed at south end
FT-02	2	300	13.41	stream channel bed at north end
FT-03	5	150	15.02	stream channel top of wall
FT-04	390	0	14.95	11 kV cable marker, south (depth 1.0 m per utility)
FT-05	390	300	15.03	11 kV cable marker, north
FT-06	300	20	15.24	old foundation corner SW
FT-07	360	60	15.27	old foundation corner NE
FT-08	20	280	15.06	meteorological mast base
FT-09	120	250	15.01	groundwater well GW-1, top of casing 15.52
FT-10	250	150	15.00	groundwater well GW-2, top of casing 15.49
FT-11	340	40	15.22	groundwater well GW-3, top of casing 15.71
FT-12	200	-60	1.10	shoreline (mean sea level datum +0.0, 2026-10 survey)
FT-13	405	150	15.12	access road edge at plant gate

- Stream channel on the west boundary: the 10 m water authority buffer (E 0-20) stays free of permanent structures; the channel bed falls from +13.41 m (north) to +13.02 m (south).
- 11 kV utility cable along the east boundary (E 390): no excavation within 2 m until its relocation by the Owner (due 2027-06-30); hand digging to locate before any crossing.
- Abandoned warehouse foundations (E 300-360, N 20-60) stand up to 0.25 m above the platform: demolition and removal by the Contractor before the platform trimming in that area.
- Meteorological mast (E 20, N 280) and groundwater wells GW-1 to GW-3 are kept; wells in conflict with foundations are replaced by the Contractor.

6. Setting-out rules

- All setting out from the control pillars CP-01, CP-02, CP-03, CP-04 and the benchmarks; pillars are checked against each other before each major setting-out and at least monthly (tolerance 10 mm plan, 5 mm height).
- Setting-out tolerances per ISO 4463-1: main grid lines ± 10 mm, foundation axes ± 5 mm, anchor bolt groups ± 3 mm (template), levels ± 5 mm.
- Every structure is set out from the plant grid; coordinates are taken from the IFC plot plan and the 3D model (never scaled from drawings).
- Construction benchmarks inside each CWA are levelled from BM-01 / BM-02 with loop closure ≤ 3 mm.

7. Findings

- Control network: transformation residuals 11 mm maximum (limit 20 mm) - the plant grid is fixed and can be used for setting out.
- Owner's platform: mean -66 mm against +15.00 m, 51 of 208 points outside ± 0.10 m, with a systematic fall towards the south and the old warehouse foundations above the platform. Final trimming to +15.00 m needs about 9,324 m³ of structural fill and 747 m³ of cut.
- TQ-039 (open): Owner platform levels below +15.00 m (survey ALP-SUR-TOP-001)

8. Site plan

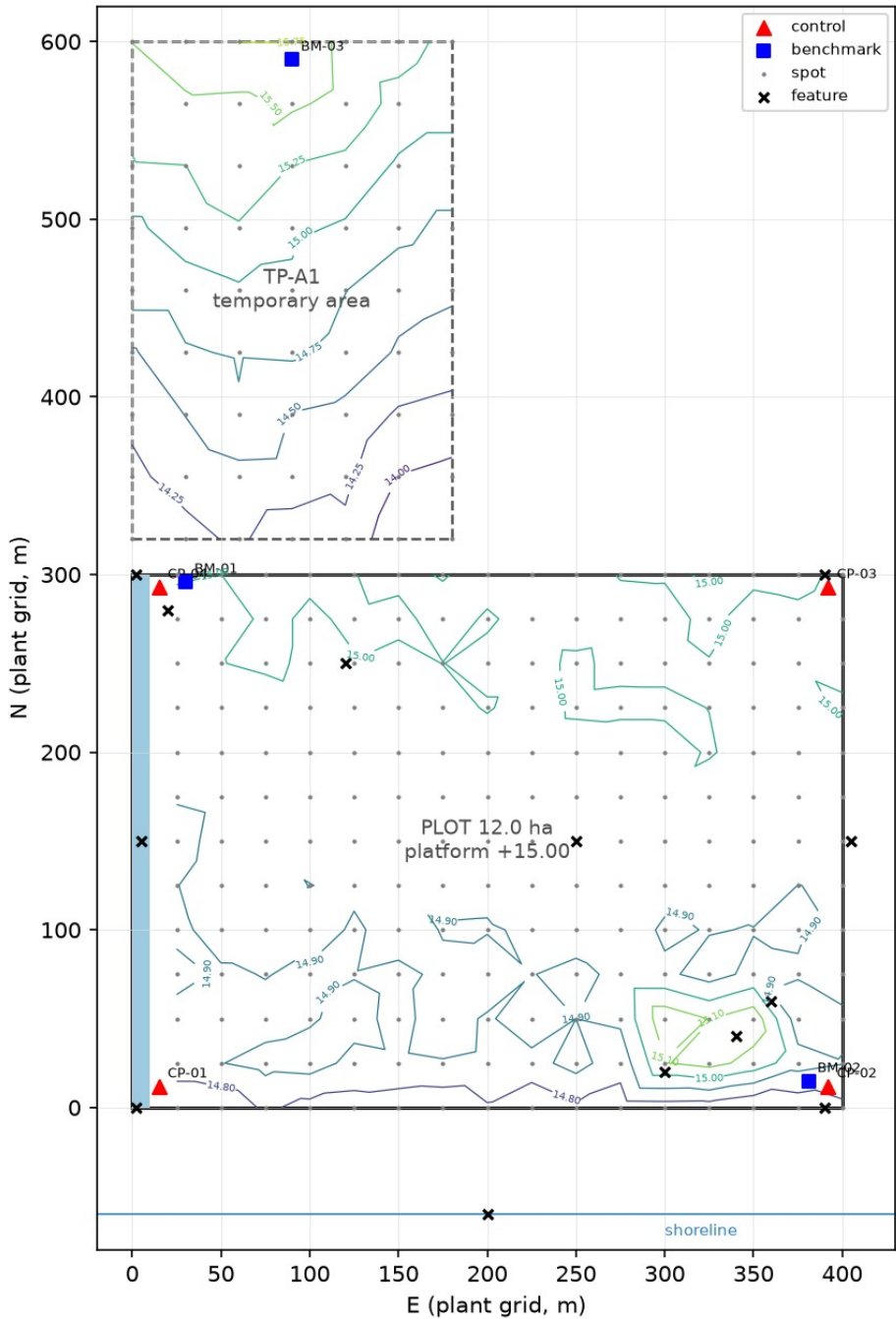


Figure 1 - Survey points and platform levels (plant grid, m); contours of the spot heights at 0.1 m (plot) and 0.25 m (TP-A1)